

Lecture (7)

Machine to Machine Communication

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Machine to Machine Communication

- ❑ M2M Communication
- ❑ M2M Service Platform (M2SP)
- ❑ Interoperability in Internet of Things

M2M Communication

❑ Machine-to-Machine (M2M)

- communication refers to the exchange of information between devices or machines equipped with computing and communication capabilities, without requiring human intervention.

Features of M2M Communication

- **Features of M2M Communication**

- **Large Number of Nodes or Devices:** Supports extensive device connectivity within the network.
- **Low Cost:** Designed to be cost-effective, making it suitable for large-scale implementations.
- **Energy Efficiency:** Devices consume minimal power, ensuring long operational life.
- **Small Traffic Per Device:** Each device transmits a relatively small amount of data.
- **Large Quantity of Collective Data:** The aggregated data from all devices is significant and requires robust handling.
- **No Human Intervention:** Machines communicate directly without the need for human input.

M2M Ecosystem

M2M Ecosystem

□ The M2M ecosystem includes several key components working together to enable seamless communication and data exchange:

- **Device Providers:**

1. Responsible for the creation and deployment of devices involved in the M2M network.
2. These devices collect and transmit data.

- **M2M Area Network:**

1. Transfers data from M2M devices through a gateway to the Internet.
2. Operated by Internet Service Providers (ISPs).

- **RESTful Architecture:**

1. Acts as an interface between device providers and ISPs.
2. Particularly suited for low-resource environments.

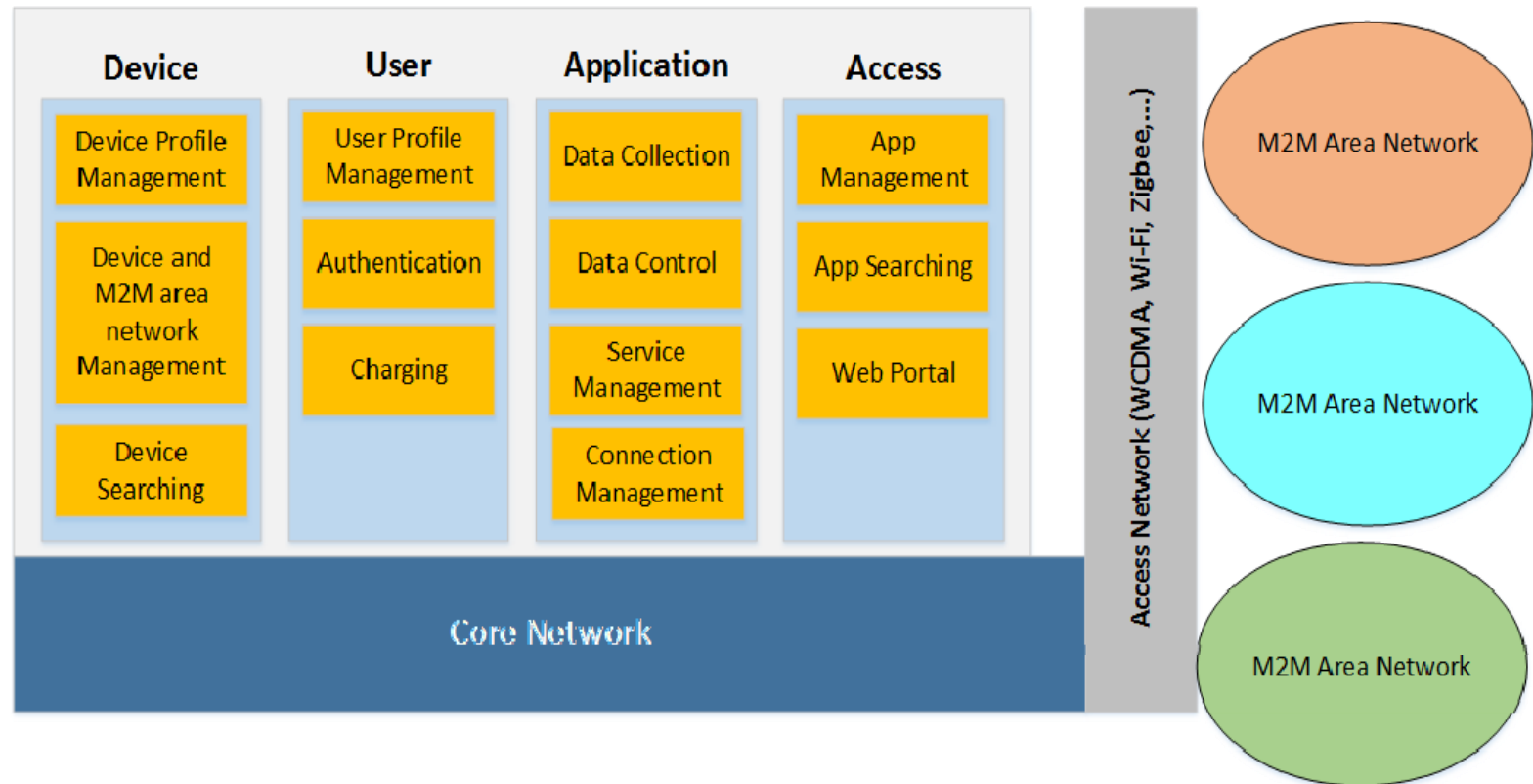
M2M Ecosystem

- **Internet Service Providers (ISPs):**
 1. Handle the transmission of data across the Internet.
- **Platform Providers:**
 1. Perform key functions such as:
 1. **Device Management:** Monitoring and maintenance of devices.
 2. **User Management:** Managing access control for users.
 3. **Data Analytics:** Processing and analyzing the collected data.
 4. **User Access:** Providing access to data for service users.
- **Service Providers and Users:**
 1. Receive and utilize the processed data for business and operational needs.
 2. RESTful architecture facilitates the interface between the platform and service providers.

M2M Platforms

- ❑ M2M platforms form the backbone of machine-to-machine communication, offering infrastructure for managing devices, users, applications, and access.
- ❑ They streamline the operations and integration of M2M systems, ensuring efficient data exchange and control.
- ❑ The platforms can be classified into the **Device Platform**, **User Platform**, **Application Platform**, and **Access Platform**.

M2M Service Platform (M2SP)



M2M Device Platform

- ❑ The M2M Device Platform focuses on managing connected devices and ensuring seamless communication.
- ❑ Key features include:
 - **Universal Access:**
 - Allows devices to be accessed anytime and anywhere via the Internet.
 - **Device Registration:**
 - Creates a comprehensive database of registered devices, enabling managers, users, and services to retrieve device information effortlessly.

M2M Device Platform

- **Profile Management:**

- Maintains device profiles, including:
 - Location
 - Device type
 - Address
 - Description

- **Authentication & Authorization:**

- Manages secure access through key authentication and authorization mechanisms.

- **Monitoring & Control:**

- Tracks the status of devices and M2M area networks.
- Provides control capabilities based on real-time status updates.

M2M User Platform

- ❑ The M2M User Platform is designed to manage users and their interactions with devices and services. Features include:
 - **User Profile Management:**
 - Handles tasks such as:
 - User registration
 - Profile modification
 - Billing/charging
 - Inquiries

M2M User Platform

- **Integration with Device Platform:**
 - Ensures interoperability by managing user access to devices, networks, and services.
- **Administrative Privileges:**
 - Grants service providers and device managers administrative control over their respective devices or networks.
- **Device Management:**
 - Enables administrators to monitor and control devices effectively.

M2M Application Platform

- ❑ This platform integrates data collected by devices and transforms it into actionable services. Key features include:
 - **Integrated Services:**
 - Combines heterogeneous datasets from various devices to create new, innovative services.
 - **Log Data Collection:**
 - Gathers logs of control processing for device management by interacting with the Device Platform.
 - **Seamless Connection Management:**
 - Manages network connections to ensure uninterrupted services.

M2M Access Platform

- ❑ The M2M Access Platform provides an interface for users to interact with M2M services via apps or web portals. Features include:
 - **Access Environment:**
 - Offers user-friendly app or web environments for accessing M2M devices and services.
 - **Service Redirection:**
 - Apps and links redirect users to the appropriate service providers.
 - **Service Delivery:**
 - Acts as the intermediary for delivering services to M2M devices.
 - **App Management:**
 - Manages app registrations by developers.
 - Provides a mapping relationship between apps and devices.
 - **Device-to-App Mapping:**
 - Lists appropriate apps for specific devices, ensuring seamless functionality.

Interoperability in Internet of Things

❑ Interoperability in Internet of Things (IoT)

- **Interoperability** is the ability of a product or system, with clearly defined interfaces, to function effectively with other products or systems—whether existing or future—without limitations in implementation or access.
- It is a cornerstone of IoT's functionality and integration.

Need for Interoperability in IoT

Need for Interoperability in IoT

- Interoperability is essential to achieve IoT objectives and address heterogeneity within the ecosystem.

❑ To Fulfill IoT Interoperability

- **Interaction and Information Sharing:** Physical objects can interact with one another and share information seamlessly.
- **Universal Communication:** Devices can communicate anytime and anywhere.
- **Communication Types:**
 - Machine-to-Machine (M2M)
 - Device-to-Device (D2D)
 - Device-to-Machine (D2M)
- **Seamless Integration:** Devices integrate smoothly with IoT networks.

To Address Heterogeneity

- ❑ IoT involves diverse technologies, protocols, and platforms, such as:
 - **Wireless Communication Protocols:** ZigBee (IEEE 802.15.4), Bluetooth (IEEE 802.15.1), GPRS, 6LoWPAN, and Wi-Fi (IEEE 802.11).
 - **Wired Communication Protocols:** Ethernet (IEEE 802.3) and Higher Layer LAN Protocols (IEEE 802.1).
 - **Programming Languages:** JavaScript, Java, C, C++, Visual Basic, PHP, Python.
 - **Hardware Platforms:** Crossbow, NI, etc.
 - **Operating Systems:**
 - For sensor nodes: TinyOS, SOS, Mantis OS, RETOS, and vendor-specific OS.
 - For personal computers: Windows, Mac, Unix, Ubuntu.

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- **Databases:** DB2, MySQL, Oracle, PostgreSQL, SQLite, SQL Server, and Sybase.
 - **Data Representations:** Variations in formats and structures.
 - **Control Models:** Variations in management and governance frameworks.
 - **Syntactic/Semantic Interpretations:** Ensuring messages are understandable in both structure and meaning.

Types of Interoperability

1. User Interoperability

Addresses the interaction challenges between users and devices. Key challenges include:

- **Device Identification and Categorization for Discovery:**
Unique addresses like Electronic Product Codes (EPC), Universal Product Code (UPC), Uniform Resource Identifier (URI), and IPv6 ensure precise device identification.
- **Syntactic Interoperability for Device Interaction:**
 - Ensures the **message formats** from devices to users are understandable by user systems.
 - Ensures the **message formats** from users to devices are executable by devices.
- **Semantic Interoperability for Device Interaction:**
 - Ensures the **meaning** of messages is mutually understandable.
 - Devices comprehend user instructions and users interpret device responses correctly.

Types of Interoperability

2. Device Interoperability

- ❑ Focuses on communication challenges between different devices.
- ❑ **Solution: Universal Middleware Bridge (UMB)**
UMB overcomes the seamless interoperability challenges caused by middleware heterogeneity.
- **How UMB Works:**
 - Creates **virtual maps** among physical devices across middleware home networks, such as HAVI, Jini, LonWorks, and UPnP.
 - Ensures compatibility among these middleware networks.

Types of Interoperability

- **Components of Universal Middleware Bridge :**
 - **UMB Core (UMB-C):**
 - Routes universal metadata messages using the Middleware Routing Table (MRT).
 - **UMB Adaptor (UMB-A):**
 - Converts physical devices into virtually abstracted devices using the Universal Device Template (UDT).
- **Universal Device Template (UDT):**

UDT provides global identifiers for device functionalities:

 - **Global Device ID**
 - **Global Function ID**
 - **Global Action ID**
 - **Global Event ID**
 - **Global Parameters**
- **Message Translation:**

UMB Adaptors translate local middleware messages into global metadata messages, ensuring seamless communication.

Any Question